

## CHADRON, NE, USA

WMO: 725636

Lat: 42.837N

Lon: 103.098W

Elev: 3294

StdP: 13.03

Time Zone: -7.00 (NAM)

Period: 00-19

WBAN: 24017

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | -8.1       | -1.8 | -14.4                         | 2.8 | -7.7 | -8.2 | 4.0 | -1.4 | 31.3                  | 27.7 | 27.5 | 31.0 | 5.0                   | 250  | 0.645 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 30.6                   | 99.4            | 66.2 | 95.7 | 65.4 | 92.3 | 65.1 | 71.0                | 87.7 | 69.3 | 86.7 | 68.0 | 85.5 | 11.6                 | 180  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 66.0                            | 108.4 | 77.5 | 64.1 | 101.3 | 75.6 | 62.6 | 96.0 | 74.9 | 36.8          | 87.1 | 35.4 | 86.7 | 34.1 | 85.2 |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |       |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|-------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |       |     |
| (4) 28.4          | 24.9 | 21.2 | DB                         | -19.8 | 104.6              | 4.5 | 3.5                                                | -23.1 | 107.1      | -25.7 | 109.1      | -28.3 | 111.1      | -31.5 | 113.6 | (4) |
| (5)               |      |      | WB                         | -20.2 | 73.9               | 4.4 | 2.2                                                | -23.4 | 75.5       | -26.0 | 76.8       | -28.5 | 78.0       | -31.7 | 79.6  | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 48.5   | 25.8  | 26.2  | 38.0  | 46.4  | 55.4  | 67.3  | 75.8  | 73.0  | 63.1  | 48.4  | 36.3  | 24.7  |
|                                                                                  | DBStd       | 20.37  | 11.42 | 12.72 | 11.54 | 9.54  | 8.97  | 7.70  | 6.18  | 6.67  | 8.79  | 9.61  | 10.81 | 12.01 |
|                                                                                  | HDD50       | 3391   | 751   | 668   | 394   | 173   | 49    | 0     | 0     | 0     | 8     | 143   | 420   | 785   |
|                                                                                  | HDD65       | 6861   | 1216  | 1087  | 838   | 557   | 317   | 61    | 4     | 15    | 137   | 518   | 861   | 1250  |
|                                                                                  | CDD50       | 2843   | 0     | 1     | 21    | 67    | 216   | 520   | 801   | 714   | 402   | 93    | 8     | 0     |
|                                                                                  | CDD65       | 840    | 0     | 0     | 0     | 1     | 19    | 131   | 340   | 264   | 82    | 3     | 0     | 0     |
|                                                                                  | CDH74       | 12373  | 0     | 0     | 29    | 125   | 434   | 1962  | 4607  | 3527  | 1504  | 179   | 6     | 0     |
|                                                                                  | CDH80       | 6215   | 0     | 0     | 2     | 20    | 140   | 901   | 2563  | 1852  | 691   | 46    | 0     | 0     |
| Wind                                                                             | WSAvg       | 8.9    | 8.7   | 8.7   | 9.4   | 10.6  | 9.7   | 9.0   | 8.6   | 8.5   | 8.7   | 8.8   | 8.4   | 8.2   |
| Precipitation                                                                    | PrecAvg     | 17.1   | 0.3   | 0.5   | 1.1   | 2.1   | 3.1   | 3.0   | 2.1   | 1.4   | 1.5   | 1.4   | 0.6   | 0.4   |
|                                                                                  | PrecMax     | 23.3   | 0.8   | 1.7   | 2.6   | 4.0   | 7.5   | 6.5   | 4.8   | 4.1   | 4.8   | 3.2   | 1.1   | 1.1   |
|                                                                                  | PrecMin     | 10.1   | 0.1   | 0.0   | 0.2   | 0.4   | 1.0   | 0.4   | 0.3   | 0.0   | 0.3   | 0.1   | 0.1   | 0.0   |
|                                                                                  | PrecStd     | 3.2    | 0.2   | 0.4   | 0.7   | 0.9   | 1.7   | 1.5   | 1.0   | 0.9   | 1.0   | 0.9   | 0.3   | 0.2   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 59.5  | 65.5  | 78.5  | 82.3  | 90.0  | 100.4 | 104.9 | 101.5 | 97.1  | 85.7  | 74.0  | 60.0  |
|                                                                                  |             | MCWB   | 42.9  | 47.7  | 52.2  | 55.1  | 63.2  | 65.7  | 67.5  | 63.8  | 64.2  | 56.8  | 51.0  | 43.3  |
|                                                                                  | 2%          | DB     | 52.5  | 57.0  | 71.4  | 77.9  | 83.7  | 94.2  | 100.4 | 97.8  | 92.3  | 79.3  | 66.3  | 53.1  |
|                                                                                  |             | MCWB   | 40.6  | 42.7  | 49.8  | 54.0  | 61.5  | 65.1  | 67.0  | 65.6  | 62.0  | 55.3  | 47.6  | 40.7  |
|                                                                                  | 5%          | DB     | 46.4  | 50.3  | 65.0  | 72.0  | 78.8  | 89.7  | 96.9  | 94.4  | 88.3  | 73.1  | 60.9  | 46.6  |
|                                                                                  |             | MCWB   | 37.3  | 39.1  | 47.5  | 52.5  | 59.6  | 65.6  | 66.4  | 65.1  | 61.0  | 53.3  | 45.2  | 37.3  |
|                                                                                  | 10%         | DB     | 42.2  | 44.8  | 58.2  | 65.6  | 73.6  | 84.8  | 92.8  | 90.6  | 83.2  | 67.2  | 54.8  | 41.5  |
|                                                                                  |             | MCWB   | 35.3  | 36.0  | 43.9  | 49.8  | 57.5  | 64.6  | 66.3  | 65.0  | 59.9  | 51.1  | 42.3  | 34.4  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 44.2  | 47.9  | 54.1  | 58.4  | 67.4  | 73.2  | 73.1  | 71.5  | 68.9  | 59.9  | 51.7  | 45.2  |
|                                                                                  |             | MCDB   | 56.3  | 65.5  | 73.4  | 76.3  | 82.8  | 86.5  | 90.4  | 89.1  | 86.9  | 78.4  | 71.4  | 58.7  |
|                                                                                  | 2%          | WB     | 40.9  | 43.1  | 50.4  | 55.5  | 63.6  | 70.1  | 71.2  | 69.6  | 65.5  | 57.0  | 48.4  | 41.0  |
|                                                                                  |             | MCDB   | 51.8  | 56.0  | 69.1  | 74.0  | 78.7  | 84.8  | 88.5  | 87.9  | 82.4  | 74.4  | 65.0  | 51.9  |
|                                                                                  | 5%          | WB     | 37.9  | 39.7  | 47.8  | 53.0  | 60.9  | 67.8  | 69.9  | 68.1  | 63.6  | 54.4  | 45.7  | 37.7  |
|                                                                                  |             | MCDB   | 45.8  | 49.1  | 63.9  | 70.4  | 74.8  | 83.6  | 87.4  | 85.8  | 82.3  | 69.7  | 60.4  | 46.4  |
|                                                                                  | 10%         | WB     | 35.4  | 36.4  | 44.9  | 50.6  | 58.7  | 65.9  | 68.5  | 66.6  | 61.7  | 52.0  | 42.7  | 34.7  |
|                                                                                  |             | MCDB   | 41.7  | 44.0  | 57.6  | 65.1  | 71.9  | 81.8  | 86.4  | 83.6  | 79.3  | 65.7  | 53.8  | 41.2  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 23.9  | 24.8  | 27.3  | 26.4  | 26.2  | 28.7  | 30.6  | 30.8  | 31.5  | 28.7  | 27.9  | 24.1  |
|                                                                                  |             | MCDBR  | 29.9  | 32.4  | 39.2  | 38.5  | 35.4  | 35.9  | 36.3  | 35.8  | 39.1  | 40.0  | 38.0  | 30.8  |
|                                                                                  |             | MCWBR  | 19.7  | 19.5  | 20.9  | 18.6  | 16.4  | 14.1  | 11.4  | 11.4  | 14.7  | 19.4  | 21.4  | 20.4  |
|                                                                                  | 5% WB       | MCDBR  | 28.0  | 30.7  | 36.8  | 35.6  | 30.6  | 30.6  | 30.6  | 30.4  | 32.8  | 35.2  | 36.6  | 30.1  |
|                                                                                  |             | MCWBR  | 19.2  | 19.2  | 20.2  | 18.1  | 15.3  | 14.5  | 11.7  | 11.2  | 13.1  | 17.6  | 20.9  | 20.4  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.256  | 0.264 | 0.285 | 0.314 | 0.336 | 0.349 | 0.372 | 0.369 | 0.327 | 0.293 | 0.268 | 0.255 |       |
|                                                                                  | taud        | 2.572  | 2.547 | 2.526 | 2.440 | 2.448 | 2.431 | 2.401 | 2.390 | 2.490 | 2.560 | 2.552 | 2.557 |       |
|                                                                                  | Ebn at Noon | 290    | 304   | 307   | 303   | 297   | 292   | 284   | 281   | 287   | 288   | 281   | 279   |       |
|                                                                                  | Edh at Noon | 21     | 25    | 29    | 34    | 35    | 36    | 36    | 36    | 30    | 24    | 21    | 19    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 663    | 948   | 1328  | 1643  | 1937  | 2279  | 2272  | 1963  | 1574  | 1060  | 731   | 555   |       |
|                                                                                  | RadStd      | 36     | 55    | 102   | 95    | 161   | 148   | 79    | 72    | 89    | 96    | 44    | 37    |       |

## Historical Trends

|                              |  | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (14 neighbors) |  | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page